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Subject : SC612

ASSIGNMENT – 1

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	P	Q	R	$(P \rightarrow Q)$	$(Q \rightarrow R)$	$(P \vee Q)$	$(P \wedge Q)$
Q-1	T	T	T	T	F	T	T
	T	T	F	F	T	T	T
	T	F	T	F	T	F	T
	T	F	F	F	F	F	F
	F	F	F	T	F	T	F
	F	T	F	F	T	T	T
	F	F	T	T	T	F	T
	F	T	T	F	F	T	F

$((P \rightarrow Q) \vee (Q \rightarrow R)) \wedge (P \vee Q)$

T
T
F
F
T
T
F
T

P₂ ②

Q-2 P = It rains.
Q = Ground is wet.

∴ $P \leftrightarrow Q$ $(\sim P \rightarrow \sim Q)$

* Contrapositive: - If the ground is not wet then it does not rain.
 $(\sim P \rightarrow \sim Q)$

* Inverse: - If it does not rain, then ground is not wet.
 $(\sim P \rightarrow \sim Q)$

* Opposite: - It rains and the ground is wet.

Q-4

P	Q	R
T	T	T

~~P~~ $[P \wedge Q \wedge \sim R] = [F]$

③

Q-3 $(P \vee Q) \wedge (\neg P \vee R) \rightarrow (Q \vee R)$

is tautology, we can prove it by logical equivalence

* Proof.

① Rewrite the implication.

$A \rightarrow B$ is logically equivalent to $\neg A \vee B$ So,

$$(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r) \equiv \neg(p \vee q)$$

$$\vee ((p \vee q) \wedge (\neg p \vee r) \wedge \neg(q \vee r))$$

* Using De Morgan's Law.

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(\neg p \vee r) \equiv \neg \neg p \wedge \neg r \equiv p \wedge \neg r$$

* So, $(\neg p \wedge \neg q) \vee (p \wedge \neg r) \vee (q \vee r)$

If $q \vee r$ is true then whole exp. is true.

If q is false then $\neg q$ is true.

If r is false then $\neg r$ is true.

* Hence Proved.

(5)

M	T	W	T	F	S	S
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$$Q-5 \quad P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

P	Q	R	$Q \vee R$	$P \wedge (Q \vee R)$	$(P \wedge Q)$	$(P \wedge R)$	$(P \wedge Q) \vee (P \wedge R)$
T	T	T	T	T	T	T	T
T	F	T	T	T	F	T	T
T	T	F	T	T	T	F	T
T	F	F	F	F	F	F	F
F	F	F	F	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	T	T	T	F	F	F	F

$$\text{Hence } P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$